QA Test Plan – AI Agent Organization Platform (MVP)

Document Purpose

This document enables QA to **plan, design, and execute testing** for the MVP of the AI Agent Organization Platform, covering functional correctness, agent behavior, governance, and system reliability.

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1. Scope of QA

In Scope

- NEW Organization mode (primary)
- EXISTING Organization mode (read-only + suggestions)
- CEO, COO, PM agents
- Artifact lifecycle
- Approval & governance workflows
- Event-driven orchestration
- UI workflows & user journeys
- Auditability & observability

Out of Scope

- Third-party integrations (Jira, GitHub write access)
- Financial transactions
- Production infrastructure deployments
- Multi-tenant scaling
- Autonomous learning

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2. Test Objectives

QA must validate that:

1. Agents operate **only within their defined authority**
2. All agent outputs are **structured artifacts**, not free text
3. Human approval gates are **mandatory and enforced**
4. End-to-end organizational flow works without manual intervention
5. System behavior is explainable, auditable, and deterministic

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3. Test Strategy

Testing Levels

- **Unit Testing**: Artifact validation, rules enforcement
- **Integration Testing**: Agent-to-agent flows via artifacts
- **System Testing**: End-to-end org scenarios
- **Exploratory Testing**: Edge cases, misuse attempts
- **Non-Functional Testing**: Performance, reliability, UX clarity

Test Environment

- Staging environment
- Seeded sample organization data
- Fixed LLM version & prompts

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4. Test Personas

Human Admin

- Creates org
- Reviews and approves artifacts
- Acts as final authority

Observer User

- Views artifacts and timelines
- No approval rights

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5. Core User Journeys to Test

Journey 1: New Organization Setup

- Create org
- Select NEW mode
- Define constraints

Expected: Org created successfully, mode locked

Journey 2: CEO Agent – Strategy Creation

Trigger: Org setup complete

Validations:

- Vision generated
- Strategic bets ≤ 3
- OKRs count between 3–5
- Artifact saved with status = Draft
- No task or execution artifacts created

Journey 3: COO Agent – Operations Design

Trigger: CEO artifact approved

Validations:

- Ops model generated
- Risk register contains ≥ 3 risks
- Delivery cadence defined

- Cannot alter CEO strategy

Journey 4: PM Agent – Product Planning

Trigger: CEO + COO artifacts approved

Validations:

- Problem statements generated
- Roadmap contains Now / Next / Later
- Sprint plan includes acceptance criteria
- Dependencies referenced correctly

Journey 5: Approval Workflow

Actions:

- Approve
- Reject with comments
- Edit & resubmit

Validations:

- Rejected artifacts cannot progress
- Approved artifacts trigger next agent
- Audit log updated

Journey 6: End-to-End Flow

Scenario:

Idea → Strategy → Ops → Roadmap → Sprint → Readiness → Final Directive

Expected:

- No skipped steps
- All approvals enforced
- Timeline view accurate

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6. Agent-Specific Test Cases

6.1 CEO Agent

Positive Tests

- Generates coherent vision
- Produces OKRs aligned with vision

Negative Tests

- Attempt to create tasks → blocked
- Attempt to bypass approval → blocked

6.2 COO Agent

Positive Tests

- Detects capacity mismatch
- Produces risk register

Negative Tests

- Attempt to redefine strategy → blocked

6.3 PM Agent

Positive Tests

- Roadmap aligns with OKRs
- Sprint scope realistic

Negative Tests

- Attempts scope outside strategy → flagged

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7. Artifact Validation

QA must validate:

- Schema correctness

- Versioning
- Status transitions (Draft → Approved → Rejected)
- Dependency integrity

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8. Governance & Security Testing

Authority Boundaries

- CEO cannot execute
- COO cannot approve
- PM cannot override strategy

Kill Switch

- Disable agents mid-flow
- Verify no further execution

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9. Observability & Audit Testing

Validate presence of:

- Decision timestamps
- Input snapshot
- Agent identity
- Confidence indicators

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10. Non-Functional Testing

Performance

- Agent execution < acceptable threshold
- UI response time under load

Reliability

- Agent retry on failure
- Graceful error handling

UX Clarity

- Clear artifact ownership
- Clear approval status

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11. Test Data Management

- Seeded example orgs
- Fixed prompts for determinism
- Resettable environment

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12. Entry & Exit Criteria

Entry Criteria

- PRD approved
- Agent charters finalized
- Test environment ready

Exit Criteria

- All critical journeys pass
- No authority violations
- No skipped approval paths
- Demo scenario passes end-to-end

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13. Risks & Focus Areas for QA

Risk	QA Focus
Agent overreach	Permission testing

Risk	QA Focus
Skipped approvals	Workflow validation
Hallucinated authority	Negative tests
Inconsistent outputs	Replay tests

14. Deliverables from QA

- Test plan (this doc)
- Test cases & scenarios
- Bug reports
- Test summary report
- Go/No-Go recommendation

15. One-Line QA Mission

> Ensure agents behave like governed executives, not uncontrolled bots — and that humans remain in control at all times.